ENHANCED DIABETES PREDICTION THROUGH MACHINE LEARNING AND ONTOLOGY

Report submitted to the SASTRA Deemed to be University
as the requirement for the course

CSE298-MINI PROJECT

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This is to certify that the project report titled “**Enhanced Diabetes Through Machine Learning and Ontology**” submitted as requirement for the course, **CSE298-MINI PROJECT** for M.C.A is a Bona-fide record of the work done by **MS. S. Rajeswari** (Reg. No. 125176071) and **MS. S. Rekha** (Reg. No. 125176073) during the academic year 2024-2025, in the **School of Computing**, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

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Date:

Project Viva voce held on

Examiner 1 **Examiner 2**

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LIST OF FIGURES

ABBREVATION

AI	Artificial Intelligence
SVM	Support Vector Machine
KNN	K-Nearest Neighbor
ANN	Artificial Neural Networks
GB	Gradient Boosting

NOTATION

English Symbols (in alphabetical order)

- *AUC* - Area Under Curve (for model evaluation)
- *Acc*- Accuracy (measure of correct predictions)
- *FN* - False Negative (incorrectly predicted negative cases)
- *FP* - False Positive (incorrectly predicted positive cases)
- *MSE* - Mean Squared Error (a loss function used in regression)
- *P* - Precision (measure of positive predictive accuracy)
- *R* - Recall (measure of sensitivity)
- *RF* - Random Forest (a machine learning model)
- *ROC* - Receiver Operating Characteristic (a graphical plot for evaluating model performance)
- *TN* - True Negative (correctly predicted negative cases)
- *TP* - True Positive (correctly predicted positive cases)

Abstract

Diabetes is a chronic disease that has been on the rise year after year. The challenges emerge when diabetes is not detected early and diagnosed promptly. Recently, a range of machine learning techniques, including ontology-based ML methods, have made significant strides in medical science by creating automated systems for detecting diabetes patients. This paper offers a comparative study and review of the most widely used machine learning techniques and ontology-based machine learning classification.

Various classification algorithms were examined, such as SVM, KNN, ANN, Naive Bayes, Logistic Regression, and Decision Tree. The evaluation of results was based on performance metrics like Recall, Accuracy, Precision, and F-Measure, derived from the confusion matrix. The experimental findings revealed that ontology classifiers and SVM provided the highest accuracy. The Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Artificial Neural Network (ANN), Naïve Bayes, Logistic Regression and Decision Tree.

Specific Contribution

- The contributions are data cleaning and the implementation of Machine learning algorithm Gradient Boosting and Gaussian Naïve Bayes to predict the Diabetes, technical analysis and to minimize the error.

Specific Learning

- The Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Artificial Neural Network (ANN), Naïve Bayes, Logistic Regression and Decision Tree. and additionally Gaussian Naïve Bayes and Extreme Gradient Boosting these are the models learned from machine learning and deep learning algorithms these are implemented to predicting the Diabetes

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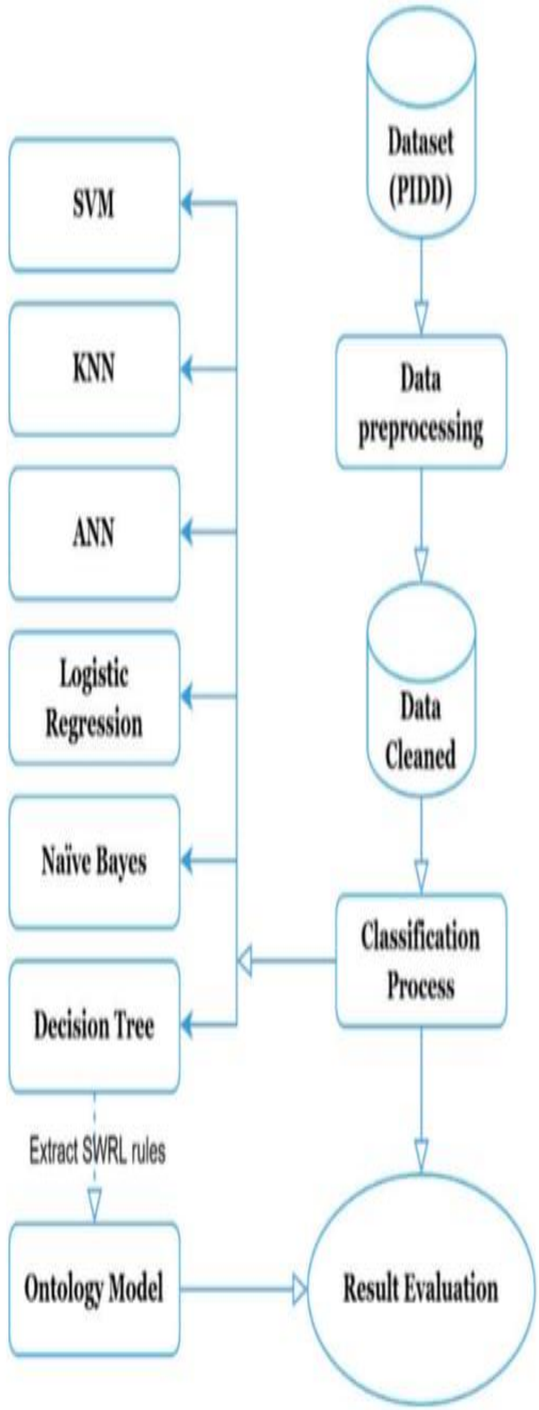
WORK FLOW OF DAIBETES PREDICTION PROJECT



CHAPTER 1

BASE PAPER SUMMARY

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ENHANCED DIABETES THROUGH MACHINE LEARNING AND ONTOLOGY

1. INTRODUCTION

The project, titled "Diabetes Prediction Using Machine Learning Algorithms and Ontology," focuses on using advanced machine learning (ML) techniques combined with ontology-based methods to improve early diabetes diagnosis. Diabetes, a chronic and growing health issue, leads to various severe complications if not diagnosed and managed promptly. The study uses popular ML classification algorithms—including SVM, KNN, ANN, Naive Bayes, Logistic Regression, and Decision Tree—to classify data and predict diabetes onset. Additionally, an ontology model was developed, leveraging relationships among health concepts to enhance the interpretability and performance of the predictions. The project analyzes these algorithms based on accuracy, precision, recall, and F-measure, using confusion matrices to evaluate their effectiveness. The findings reveal that the ontology-based classifier and SVM yield superior accuracy, suggesting that combining ML with ontological reasoning can enhance classification results. This comparative study emphasizes the potential of integrating ontologies with ML to advance predictive accuracy and decision support in healthcare.

1.1. PROBLEM STATEMENT

Diabetes is a significant global health challenge due to its chronic nature and the complications arising from late diagnosis or improper management. Early detection is crucial to preventing or minimizing severe health issues. Traditional diagnostic methods are often time-consuming and may lack the precision needed for early intervention. Machine learning (ML) methods offer promising solutions for automating the detection process, yet ML algorithms alone may not fully leverage the complex, interrelated medical data associated with diabetes. The problem this project addresses is how to improve the accuracy and interpretability of diabetes prediction models by combining traditional machine learning techniques with ontology-based classification. By integrating ML and ontology methods, this study aims to develop a more robust, automated system capable of early and accurate diabetes prediction, ultimately supporting healthcare providers in making timely, informed decisions for diabetic patient care.

1.2. LITERATURE SURVEY

1. **Alaa Khaleel & Al-Bakry (2021)** - This study focused on diabetes prediction using ML algorithms, specifically evaluating precision, recall, and F1-measure. By using the Pima Indian Diabetes Database (PIDD), the authors achieved high accuracy with Logistic Regression, Naïve Bayes, and K-Nearest Neighbor, which reported accuracies of 94%, 79%, and 69%.

2. **Khanam & Foo (2021)** - The authors applied seven ML algorithms to predict diabetes, finding Logistic Regression and SVM as the most effective. By experimenting with neural networks (NNs), they demonstrated that a two-layer NN yielded an 88.6% accuracy, underscoring the robustness of both classical algorithms and advanced neural models.

3. **Sarwar et al. (2018)** - This study applied various ML algorithms to healthcare datasets for diabetes prediction, focusing on accuracy and performance comparison. The authors highlighted the potential of predictive analytics in healthcare and provided insights into the efficiency of different algorithms in detecting diabetic patients early.

4. **Mujumdar & Vaidehi (2019)** - This research included both standard and external factors, like BMI, age, and insulin levels, for diabetes prediction. The authors noted improved classification accuracy with their expanded dataset, contributing to better diabetes prediction models that consider a broader range of contributing factors

5. **Rady et al. (2021)** - Using a dataset of 521 cases, the authors compared eight ML algorithms, with Random Forest achieving 98% accuracy. This high performance emphasized the effectiveness of ensemble methods, like Random Forest, for medical diagnosis applications, including diabetes prediction.

6. **Emon et al. (2021)** - This paper presented a model for early-stage diabetes prediction using ML, with Logistic Regression, Gaussian Process, Adaptive Boosting, and other algorithms. Random Forest emerged as the top performer, achieving 98% accuracy, supporting the utility of ensemble techniques in healthcare diagnostics.

1.3. EXISTING WORK

Numerous studies have explored ML algorithms for diabetes diagnosis and classification using standard medical datasets, particularly the Pima Indian Diabetes Database (PIDD). Algorithms like Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Artificial Neural Network (ANN), Naïve Bayes, Logistic Regression, and Decision Tree have been widely applied. Past research has shown that ML methods, such as Random Forest and Logistic Regression.

1.4. PROPOSED WORK

Gradient Boosting is recognized for its ability to enhance predictive accuracy by sequentially improving upon weak learners, typically Decision Trees, through iterative minimization of error. Gaussian Naive Bayes offers interpretability, as its probabilistic outputs allow clinicians to understand the likelihood of diabetes based on various features (e.g., probability of diabetes given a high BMI or glucose level).

2. MERITS AND DEMERITS OF BASE PAPER

Merits:

1.Comprehensive Comparison: The paper evaluates multiple ML classifiers (SVM, KNN, ANN, etc.) alongside ontology-based classifiers, providing a well-rounded perspective on diabetes prediction techniques.

2. Integration of Ontology: By integrating machine learning with ontology-based methods, the study enhances decision interpretability, making the results more actionable for medical practitioners.

3. Robust Evaluation Metrics: Use of diverse metrics (Accuracy, Precision, Recall, F-Measure, ROC-AUC) ensures a thorough performance assessment.

4. Use of Open-Source Tools: Tools like Weka and Protégé make the methodology reproducible and accessible for researchers.

5. Innovative Approach: Combining ontology with ML is a novel contribution, especially in the domain of diabetes prediction.

6. Focus on Interpretability: The ontology-based classifier provides human-readable reasoning, which aids in trust and adoption.

Demerits:

1. Limited Dataset: The study relies solely on the Pima Indians Diabetes Database, which is relatively small and may limit generalizability.

2. No Feature Selection: The absence of feature selection techniques could impact the optimization of classifier performance.

2. 1. OBJECTIVES

Evaluating and comparing the effectiveness of popular machine learning algorithms (such as SVM, KNN, ANN, Naive Bayes, Logistic Regression, and Decision Tree) and ontology-based classification models for diabetes prediction. Investigating the integration of ontology with machine learning for structured and interpretable diabetes prediction, aiming to enhance decision-making in healthcare. Assessing each model based on metrics such as accuracy, precision, recall, and F-measure to determine the most effective approach for early diabetes diagnosis.

3. EXPERIMENTAL WORK

The experimental work for the paper involved using the Pima Indians Diabetes Database (PIDD) with 768 instances and 8 attributes, which was preprocessed and formatted for use with Weka software. Six classification algorithms—SVM, KNN, ANN, Logistic Regression, Naive Bayes, and Decision Tree—were implemented to classify patients as diabetic or non-diabetic. An ontology-based classifier was also developed using the Protegé platform, where rules from a Decision Tree algorithm were embedded within the ontology to infer diabetes status through the SWRLTab plugin and Pellet reasoner. The performance of both the machine learning models and the ontology-based classifier was evaluated based on accuracy, precision, recall, and F-measure, using 10-fold cross-validation and a 66% train-test split. The comparative analysis revealed that the ontology-based classifier and SVM achieved the highest accuracy in diabetes prediction, highlighting the potential benefits of integrating machine learning with ontology for interpretable and effective classification in healthcare.

3.1. SUPPORT VECTOR MACHINE

One of the most flexible and dependable supervised machine learning techniques is the vector machine. This implies that it frequently yields excellent outcomes regardless of thecircumstance. It also applies to concerns with regression in addition to classification challenges. Figure 2 shows the result of years of work by a variety of individuals, which is SVM. The first SVM algorithm was credited to Vladimir Vapnik in 1963. In Figure 2, Support Vector Machine isdescribed.

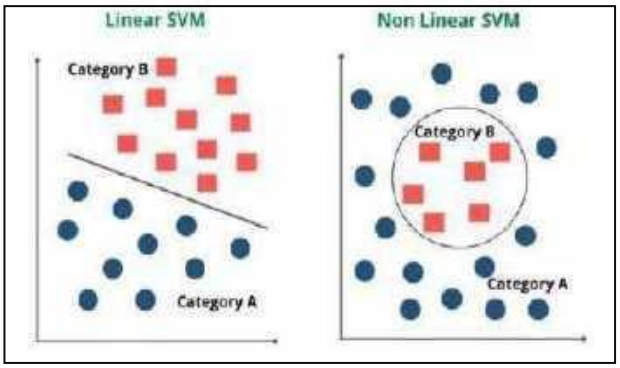


FIGURE 2: SUPPORT VECTOR MACHINE

3.2. K-NEAREST NEIGHBOR

Using a similarity measure, KNN is a straightforward algorithm that maintains all existing examples and classifies incoming data or instances. This method mainly classifies a data point based on the classification of its neighbors. Although the KNN approach generates incredibly accurate recommendations, it can outperform even the most exact mathematical models.

Thus, the KNN technique will be helpful to solve issues that don't require a human-readable model but do require a high degree of accuracy. The distance measure determines how accurate the predictions are. Figure 3 depicts the KNN,

which is one of the most widely used algorithms for text recognition. Another interesting use is the evaluation of forest inventories and the calculationof forest attributes. K Nearest Neighbor is described in Figure 3.

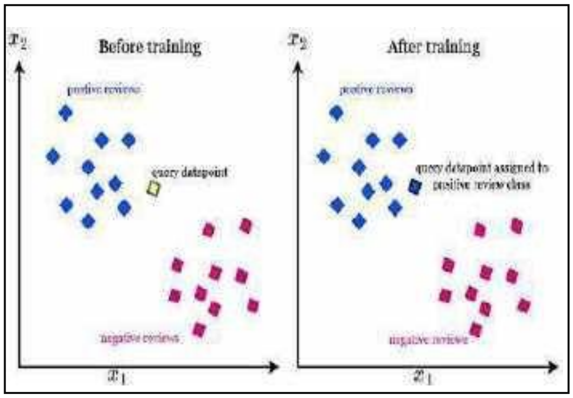


FIGURE 3: K-NEAREST NEIGHBOR

3.3. GRADIENT BOOSTING

One kind of machine learning boosting is called gradient boosting, which is predicatedon the idea that combining the best later model with earlier models can lower the overall prediction error. By defining the intended outcomes for the subsequent model, the most important objective is to minimize error. When forecasting continuous and categorical target variables, the gradient boosting technique can be applied as a classifier or regression. The Mean Square Error (MSE) cost function is utilized as a regressor, and log loss is the cost function used as a classifier. Gradient boosting is frequently used when decision trees (especially CARTs) of a fixed size are used as base learners. Figure 5 shows gradient boosting. Friedman advises modifying the gradient boosting technique to increase each base learner's degree of fit

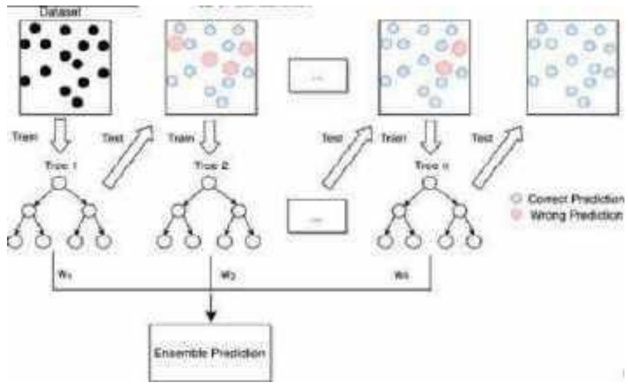


FIGURE 4: Gradient Boosting

3.4. ANN

An Artificial Neural Network (ANN) is a computational model inspired by the structure and function of the human brain. ANNs consist of interconnected layers of "neurons" (nodes) that process information in multiple layers: the input layer, one or more hidden layers, and an output layer. Each neuron in a layer receives input from neurons in the previous layer, processes it through an activation function (such as ReLU or sigmoid), and passes the result to neurons in the next layer.

The core learning mechanism in an ANN is achieved through **backpropagation** and **gradient descent**. Initially, the network is assigned random weights. As data flows through the network, the predicted output is compared to the actual output, and the difference (error) is used to adjust the weights. This process iteratively reduces the error over multiple rounds (epochs), allowing the network to learn from the data.

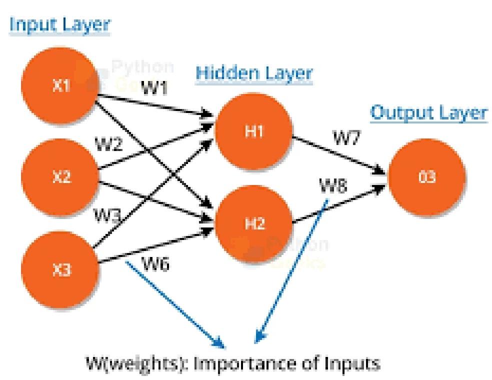


FIGURE.5 ANN

3.5. NAÏVE BAYES

Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem, which assumes that the features (or predictors) are conditionally independent given the class label. This "naive" assumption simplifies calculations and makes the algorithm highly efficient, although it may not always hold in practice. The algorithm calculates the probability of each class given the input features, and the class with the highest probability is chosen as the prediction. For each class, it estimates the likelihood of the input features by using the formula.

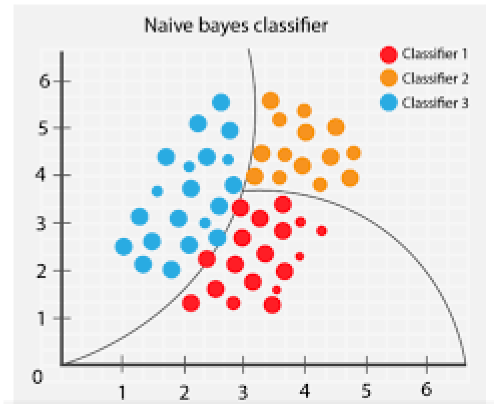


FIGURE 6. NAÏVE BAYES

3.6 DECISION TREE

A Decision Tree is a supervised learning algorithm used in both classification and regression tasks. It operates by recursively splitting data into subsets based on specific feature values, forming a tree-like structure of decisions leading to various outcomes. The process starts at the root node, where the algorithm selects a feature and a threshold to split the data, aiming to maximize homogeneity within each branch. This selection is guided by metrics like Gini impurity or information gain for classification. Each node continues to split recursively until a stopping condition is met, such as reaching a maximum depth, achieving pure nodes (where all samples belong to the same class), or having too few samples for meaningful splits. The final nodes, known as leaf nodes, contain the predictions; in classification tasks, each leaf holds the class label most frequent among samples in that node. Decision Trees are valued for their interpretability and ability to handle both categorical and numerical data without requiring feature scaling. However, they are prone to overfitting, especially with deeper trees, which can be mitigated through pruning or ensemble methods like Random Forests that improve accuracy and robustness.

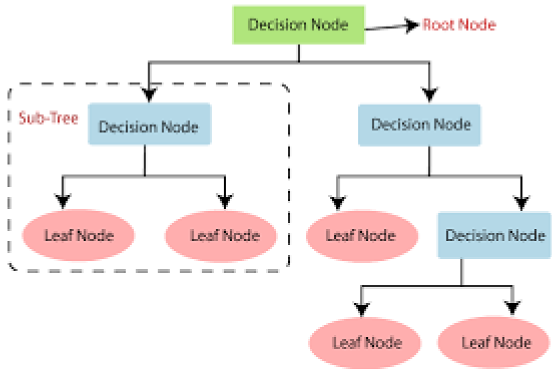


FIGURE 7: DECISION TREE

3.7 LOGISTIC REGRESSION

Logistic Regression is a supervised learning algorithm used primarily for binary classification tasks. Unlike linear regression, which predicts continuous values, logistic regression estimates the probability that a given input belongs to a particular class. The algorithm does this by applying the logistic (sigmoid) function to a linear combination of input features, mapping any real-valued number to a range between 0 and 1, representing probability.

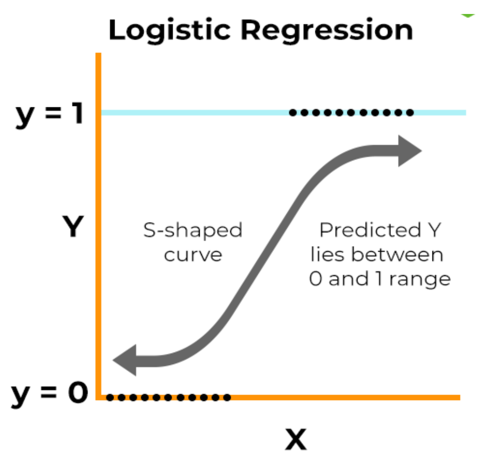


FIGURE: 8 LOGISTIC REGRESSION

3.8 GAUSSIAN NAÏVE BAYES

Gaussian Naive Bayes is a variant of the Naive Bayes classification algorithm specifically designed for continuous data. It assumes that each feature follows a normal (Gaussian) distribution. This algorithm uses Bayes' Theorem to calculate the probability of each class given the input features, making a "naive" assumption that all features are conditionally independent given the class label.

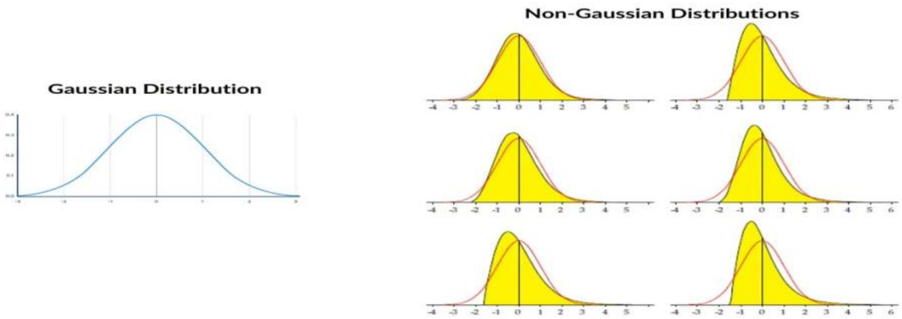


FIGURE 9: GAUSSIAN NAÏVE BAYES

3.9. DATA PRE-PROCESSING

A sequence of crucial procedures known as data preparation are employed in stock prediction to convert unprocessed stock market data into a format that is clear, standardized, and practical for the development of network models. This procedure begins by collecting info through multiple sources, then moves on to cleaning the data to address problems like duplicates, outliers, and missing numbers. By generating useful characteristics of the raw data, like window information, lagged characteristics, and technical indicators, feature design performs a critical role. By guaranteeing that the characteristics have comparable scales, normalizing or standardizing the data enhances the efficacy of training the model. To avoid lookahead bias, the information is subsequently separated into training, validation, and testing sets while keeping the sequence of events intact. To increase the exactness of models and avoid bias, further consideration is given to temporal aspects and managing incorrect information circumstances. In order to gather trustworthy thoughts and provide accurate forecasts in stock market estimations, these preparation processes become crucial.

4. CODE

```
# Importing necessary libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt
```

```
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load the dataset
dataset = pd.read_csv('diabetes.csv')

# Data preprocessing - replacing zero values with NaN for specific columns
columns_to_replace = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]
dataset[columns_to_replace] = dataset[columns_to_replace].replace(0, np.NaN)

# Fill NaN values with the mean of each column
for column in columns_to_replace:
    dataset[column].fillna(dataset[column].mean(), inplace=True)

# Feature scaling
```

```
scaler = MinMaxScaler()

dataset_scaled = scaler.fit_transform(dataset)


# Splitting features and target variable

X = dataset_scaled[:, :-1]  # All columns except the last one (Outcome)

Y = dataset_scaled[:, -1]   # Outcome column

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.20, random_state=42)
```

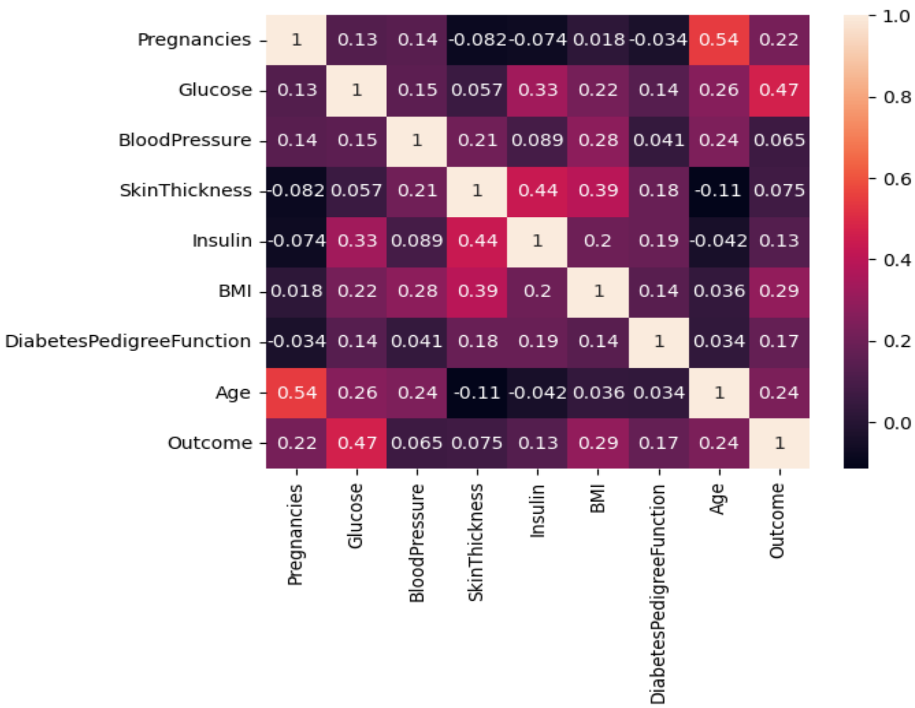


FIGURE9: CORRELATION MATRIX

```
# Initializing models

logreg = LogisticRegression(random_state=42)

knn = KNeighborsClassifier(n_neighbors=24, metric='minkowski', p=2)

svc = SVC(kernel='linear', random_state=42)
```

```
nb = GaussianNB()
dectree = DecisionTreeClassifier(criterion='entropy', random_state=42)
ranfor = RandomForestClassifier(n_estimators=11, criterion='entropy', random_state=42)

# Training models
logreg.fit(X_train, Y_train)
knn.fit(X_train, Y_train)
svc.fit(X_train, Y_train)
nb.fit(X_train, Y_train)
dectree.fit(X_train, Y_train)
ranfor.fit(X_train, Y_train)

# Making predictions on test data
Y_pred_logreg = logreg.predict(X_test)
Y_pred_knn = knn.predict(X_test)
Y_pred_svc = svc.predict(X_test)
Y_pred_nb = nb.predict(X_test)
Y_pred_dectree = dectree.predict(X_test)
Y_pred_ranfor = ranfor.predict(X_test)

# Calculating accuracy for each model
accuracy_results = {
    "Model": ["Logistic Regression", "K-Nearest Neighbors", "Support Vector Classifier",
```

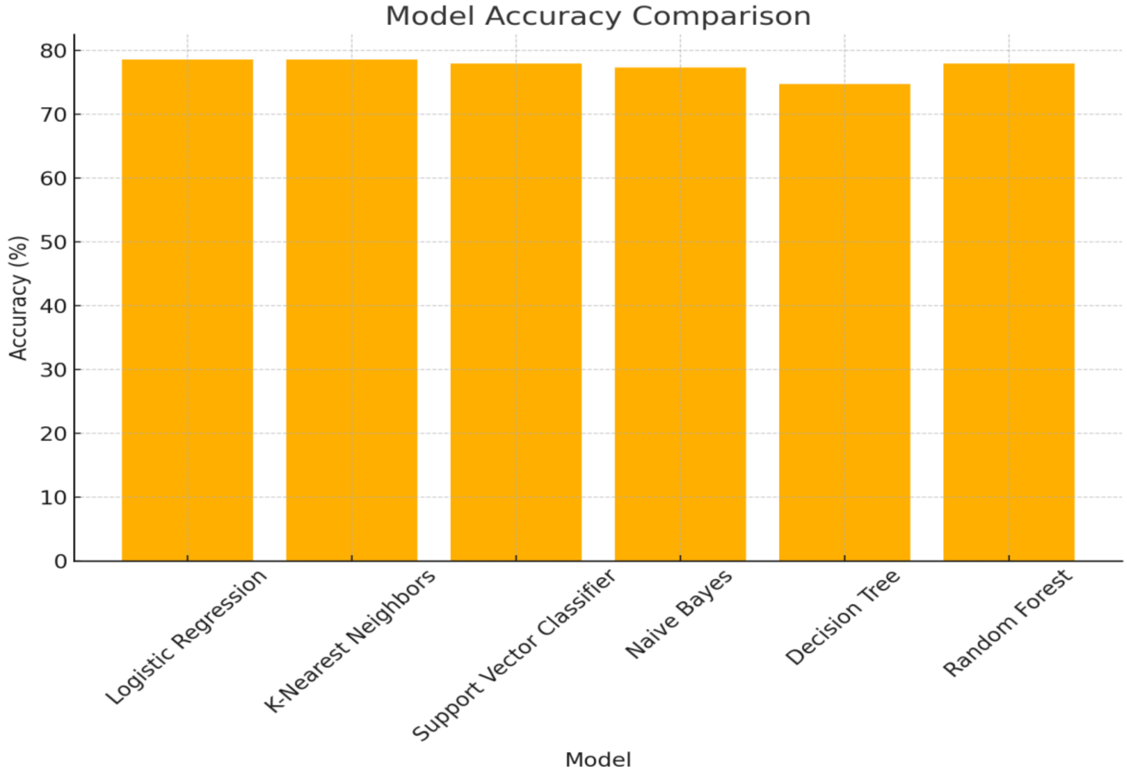
```
            "Naive Bayes", "Decision Tree", "Random Forest"],
    "Accuracy (%)": [
        accuracy_score(Y_test, Y_pred_logreg) * 100,
        accuracy_score(Y_test, Y_pred_knn) * 100,
        accuracy_score(Y_test, Y_pred_svc) * 100,
        accuracy_score(Y_test, Y_pred_nb) * 100,
        accuracy_score(Y_test, Y_pred_dectree) * 100,
        accuracy_score(Y_test, Y_pred_ranfor) * 100
    ]
}

# Converting to DataFrame
accuracy_df = pd.DataFrame(accuracy_results)
print("Model Accuracy Comparison:\n", accuracy_df)

# Plotting accuracy comparison chart
plt.figure(figsize=(10, 6))
plt.bar(accuracy_df["Model"], accuracy_df["Accuracy (%)"])
plt.title("Model Accuracy Comparison")
plt.xlabel("Model")
plt.ylabel("Accuracy (%)")
plt.xticks(rotation=45)
plt.show()
```

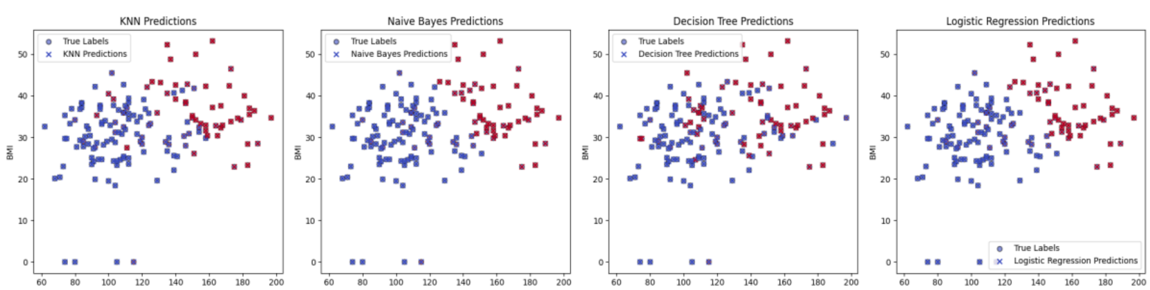
5. RESULTS

The results of the diabetes classification models show varying levels of accuracy and predictive power. Logistic Regression and K-Nearest Neighbors both achieved the highest accuracy of 78.57%, indicating these models are effective in classifying diabetic vs. non-diabetic cases in this dataset. The Support Vector Classifier and Random Forest models follow closely with an accuracy of 77.92%, suggesting strong performance but slightly less consistent than the top models. Naive Bayes achieved a reasonable accuracy of 77.27%, benefiting from its simplicity and efficiency on high-dimensional data. The Decision Tree model, while interpretable and flexible, yielded the lowest accuracy at 74.67%, likely due to its tendency to overfit on smaller datasets without pruning.Confusion matrices for each model indicate that true positives (correctly predicted diabetic cases) and true negatives (correctly predicted non-diabetic cases) were generally high, with slight variations in false positives and false negatives across models. Logistic Regression and K-Nearest Neighbors showed a balanced trade-off between sensitivity (ability to detect diabetic cases) and specificity (ability to detect non-diabetic cases), making them robust options for binary classification in this scenario.Overall, while K-Nearest Neighbors and Logistic Regression emerged as the most accurate classifiers for this dataset, models like Naive Bayes and Random Forest also demonstrated reliable performance, showcasing the effectiveness of different algorithms for medical prediction tasks.

ALGORITHM	PRECISION	RECALL	F1-SCORE	ACCURACY
LOGISTIC REGRESSION	1.0	1.0	0.95000	0.7207
SVM	0.8	0.8	0.487179	0.7337
NAÏVE BAYES	0.7	0.67	0.710145	0.7142
DECISION TREE	0.57	0.75	0.980000	0.681818
RANDOM FOREST	0.8	0.8	0.823529	0.759740
KNN	1.0	0.8	0.644068	0.7857

ACCURACY TABLE





SCATTER PLOT FOR THE PREDICTION OF ALFORITHM ON TRUE POSITIVES AND NEGATIVES

6. CONCLUSION AND FEATURE WORKS

This study compared several machine learning models—Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Naive Bayes, Decision Tree, and Random Forest—for diabetes prediction. The results indicate that Logistic Regression and KNN achieved the highest accuracy, making them the most reliable models for this dataset, with SVC and Random Forest close behind. Naive Bayes provided a competitive performance, while Decision Tree accuracy was somewhat lower due to its tendency to overfit on small datasets. These findings reinforce the effectiveness of simpler models like Logistic Regression and KNN for medical diagnosis, where interpretability, efficiency, and accuracy are critical.

Future Work:

Future efforts can focus on enhancing model accuracy and robustness by experimenting with ensemble techniques, such as boosting and bagging, which may improve Decision Tree and Random Forest performance. Additionally, applying neural networks or deep learning architectures could be explored for their potential to capture complex patterns, especially in larger and more varied medical datasets. Further, incorporating feature engineering and dimensionality reduction techniques could streamline the input data and improve accuracy. Expanding the study to include diverse populations or using more extensive datasets can also make the models more generalizable and clinically applicable.

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APPENDIX

Diabetes Prediction Using Machine Learning Algorithms and Ontology

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Abstract

Diabetes is one of the chronic diseases, which is increasing from year to year. The problems begin when diabetes is not detected at an early phase and diagnosed properly at the appropriate time. Different machine learning techniques, as well as ontology-based ML techniques, have recently played an important role in medical science by developing an automated system that can detect diabetes patients. This paper provides a comparative study and review of the most popular machine learning techniques and ontology-based Machine Learning classification. Various types of classification algorithms were considered namely: SVM, KNN, ANN, Naive Bayes, Logistic regression, and Decision Tree. The results are evaluated based on performance metrics like Recall, Accuracy, Precision, and F-Measure that are derived from the confusion matrix. The experimental results showed that the best accuracy goes for ontology classifiers and SVM.

Keywords: Machine learning, ontology, diabetes, prediction.

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1 Introduction

Diabetes is a group of the deadliest and metabolic diseases in which the level of blood sugar in the human body is abnormally high. It impacts the body's capacity to produce the hormone insulin. High blood sugar commonly causes many complications such as intensified thirst, increased hunger, and frequent urination if diabetes goes untreated and undiagnosed at an early stage. Therefore, early detection is the only way to avoid complications by using the trending technology for ontology-based and machine learning.

Machine learning (ML) is one of the most rapidly developing fields of computer science, with several applications. It refers to the process of extracting useful information from a large set of data. ML techniques are used in different areas such as medical diagnosis, marketing, industry, and other scientific fields. ML algorithms have been widely used in medical datasets and are best suited for medical data analysis. There are various forms of ML, including classification, regression, and clustering. Depending on the problem that we are trying to solve, each form has a distinct result and impact. In our work, we focus on classification methods, which are applied to classify a given dataset into predefined groups and to predict future activities or information to that data due to its good accuracy and performance. Classification algorithms are usually employed in the medical domain especially in diagnosing the diseases such as diabetes. Therefore, the commonly used machine learning classification [1] namely SVM, KNN, ANN, Naive Bayes, Logistic regression, and Decision Tree are applied to identify diabetes patients at an early period.

On the other side, Ontology is one of the most adopted approaches to manage, organize and extract data during the previous decades. It is a data representation method that has been successfully implemented in a variety of fields, especially the medical domain. It is important in computer science because of its capacity to represent diverse concepts and their relationships in different disciplines. In actuality, no single ontology is sufficient to follow the growing demands of today's healthcare, and the ontologies must be integrated with algorithms of machine learning to support data integration and analysis [2]. In previous work, we already created and explored an ontology-based model capable of predicting diabetes patients by using an ontology classifier based on a decision tree algorithm.

In this study, we aim to make a comparative analysis among the six popular classification techniques and ontology-based machine learning classification based on carefully chosen parameters such as Precision, Accuracy, F-Measure, and Recall, which are derived from the confusion matrix.

The organization of the remainder of the paper is as follows: Section 2 represents the literature review of related classification algorithms in the field of diabetes prediction. Section 3 we present technologies and methods used in this comparative analysis. Section 4 we describe the performance metrics used to evaluate the models. Section 5, we present results and discussion. Finally, Section 6 presents future work and conclusions.

2 Related Works

Recently researchers have published a considerable amount of research to identify diabetic patients based on symptoms by applying machine-learning techniques. In [3], the authors propose a model that can predict is the patient has diabetes or not. This model is based on the prediction precision of powerful machine learning algorithms, which use certain measures such as precision, recall, and F1-measure. The authors use Pima Indian Diabetes (PIDD) dataset to predict diabetic onset based on diagnostics manner. The results obtained using Logistic Regression (LR), Naïve Bayes (NB), and K-nearest Neighbor (KNN) algorithms were 94%, 79%, and 69% respectively. In the paper [4], the authors use seven ML algorithms on the dataset to predict diabetes, they found that the model with Logistic Regression and SVM were better on diabetes prediction, they built a NN model with a different hidden layer and observed the NN with two hidden layers provided 88.6% accuracy.

The study applied in the paper [5] uses several machine learning classification algorithms (Gaussian Naive Bayes, K-Nearest Neighbors, Artificial Neural Network, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine) on the PIID dataset. Logistic Regression got the best accuracy result.

Sarwar et al. [6], discuss predictive analytics in healthcare, a number of machine learning algorithms are used in this study. For experiment purposes, a dataset of patient's medical is obtained. The performance and accuracy of the applied algorithms are discussed and compared

In the paper [7], the authors propose a diabetes prediction model for the classification of diabetes including external factors responsible for diabetes along with regular factors like Glucose, BMI, Age, Insulin, etc. Classification accuracy is improved with the novel dataset compared with existing dataset.

On a dataset of 521 instances (80% and 20% for training testing respectively), [8] authors applied 8 ML algorithms such as logistic regression, support vector machines-linear, and nonlinear kernel, random forest, decision tree, adaptive boosting classifier, K-nearest neighbor, and naïve bayes. According to the results, the Random Forest classifier achieved 98% accuracy compared to the other.

In [9], the researchers used machine-learning algorithms including Logistic Regression, Gaussian Process, Adaptive Boosting (AdaBoost), Decision Tree, K-Nearest Neighbors, Multilayer Perceptron, Support Vector Machine, Bernoulli Naive Bayes, Bagging Classifier, Random Forest, and Quadratic Discriminant Analysis. The Random Forest classifier performs better and achieved a 98% accuracy, which is higher than the other three algorithms.

To predict diabetes at an early stage, the paper [10] proposes a novel approach to diabetes prediction using significant attributes. Various tools are used to determine attribute selection for clustering and prediction. The results indicate a strong association of diabetes with body mass index (BMI) and glucose level. Several techniques for predicting diabetes are used such as artificial neural network (ANN), random forest, and K-means clustering for the prediction of diabetes, and the ANN technique provided the best accuracy

Another method is used for diabetes prediction [11]. In this method the authors propose a novel approach of machine learning algorithms applied in hadoop based clusters for diabetes prediction. This approach is applied in the Pima Indians Diabetes Database and Digestive Diseases and the results obtained show that the ML algorithms produce the best accurate diabetes predictive.

In this experimental analysis [12] four machine learning algorithms, Random Forest, K-nearest neighbor, Support Vector Machine, and Linear Discriminant Analysis are used in the predictive analysis of early-stage diabetes. High accuracy of 87.66% goes to the Random Forest classifier.

In another way, the authors of the paper [13] have built models to predict and classify diabetes complications. In this work, several supervised classification algorithms were applied to predict and classify 8 diabetes complications. The complications include some parameters such as metabolic syndrome, dyslipidemia, nephropathy, diabetic foot, obesity, and retinopathy.

In [14], the authors present two approaches of machine learning to predict diabetes patients. Random forest algorithm for the classification approach, and XGBoost algorithm for a hybrid approach. The results show that XGBoost outperforms in terms of an accuracy rate of 74.10%.

Authors in this article [15] tested machine learning algorithms such as support vector machine, logistic regression, Decision Tree, Random Forest, gradient boost, K-nearest neighbor, Naïve Bayes algorithm. According to the results, Naïve Base and Random Forest classifiers achieved 80% accuracy compared to the other algorithms.

3 Technologies and Method

The experimentation is carried out using the methods and technologies described in the next subsections. The process of developing this comparative analysis is illustrated in Figure 1.

3.1 Dataset

The dataset called Pima Indians Diabetes Database (PIDD) is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. The purpose is to expect based on diagnostic measurements whether a patient has diabetes. It has 768 instances and 8 numerical attributes plus a class (preg, plas, pres, skin, insu, mass, pedi, age, class).

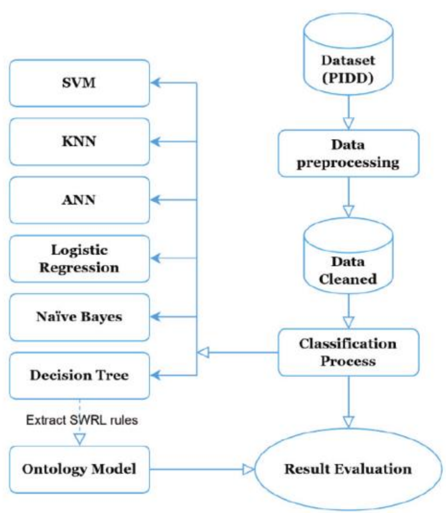

Figure 1 Experimental flowchart.

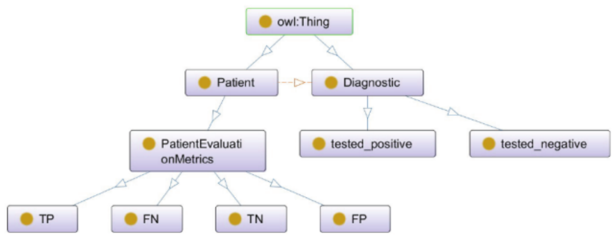

Figure 2 Graphical representation of the ontology.

After the dataset pre-processing step using UCI Machine Learning, the output file in CSV format will be transformed into ARFF format.

3.2 Machine Learning Algorithms

After preparing the dataset, we import it into Weka software, which contains tools for data preparation, classification [16], clustering, association rule exploration, visualization [17] and Similarity [18]. We used the six most commonly used classifiers to classify binary datasets (SVM, KNN, ANN, Logistic Regression, Naïve Bayes, Decision Tree). The results of the classifiers can be found in Section 5.

3.3 Ontology Model

The approach used to classify the dataset using the ontology model was published and detailed in our previous work [2], we recommend reading it for more details. Here, we will give some details briefly.

The ontology was created by the open-source platform “Protégé”, a free ontology editor and framework for building intelligent systems [19]. Figure 2 illustrates the graphical representation of our ontology generated by the OntoGraph plugin.

The dataset is imported with the help of Cellfie, a Protégé plugin for importing spreadsheet data into OWL ontologies. Then, we extracted generated rules from the Decision Tree algorithm and import them to Protégé using the SWRLTab plugin. To execute SWRL rules and infer new ontology axioms, we used the Pellet reasoner which has a more direct functionality for working with OWL and SWRL rules. It uses the dataset and SWRL rules to induce the inference and provides the final decision where is the patient is tested negative or positive. The results of the ontology classifier are presented in Section 5.

4 Evaluation

In Machine Learning, performance measurement is an essential task. It is critical to choose the right metrics to evaluate the machine learning model. Therefore, metrics are used to determine how machine learning algorithms' performance is measured and compared.

Different performance metrics are used to evaluate machine learning algorithms such as Accuracy, Precision, Recall, F-Measure, ROC Area, Kappa statistic, Root mean squared error, Root relative squared error, etc.

Almost all of the performance metrics are derived from the Confusion Matrix and the numbers inside it. The Confusion Matrix is one of the most intuitive and easiest metrics for determining the model's correctness and accuracy. It is used for classification problems with two or more types of classes as output.

The confusion matrix is a table with two dimensions ("Actual" and "Predicted"), and sets of "classes" in both dimensions. Our Actual classifications are columns and Predicted ones are Rows. For more understanding of what the confusion matrix is all about and what it represents, let's take a real example from our study where we are predicting whether a patient is having diabetes or not (1: tested positive 0: tested negative). Figure 3 illustrates the confusion Matrix details, and Table 1 describes the terms associated with the confusion matrix.

An ideal classification performance would only have no entries for FN and FP (i.e., the number of FN equal number of FP equal zero).

Diverse measures can be derived from a confusion matrix such as Accuracy, Precision, Recall and F-Measure. The best value of accuracy, precision, and recall is 1.0, whereas the worst is 0.0. Figure 3 illustrates how to compute them from the confusion matrix.

Table 1 Terms associated with Confusion matrix

Terms	Description
True Positives (TP)	Number of patients correctly identified as Positive
True Negatives (TN)	Number of patients correctly identified as Negative
False Positives (FP)	Number of patients incorrectly identified as Positive
False Negatives (FN)	Number of patients incorrectly identified as Negative

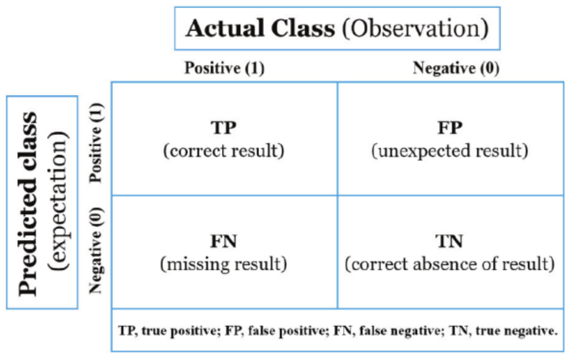

Figure 3 Confusion Matrix details.

Accuracy (ACC):

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy is computed as the number of all correct predictions divided by the total number of the dataset, which is the number of patients that are identified correctly in total in our case.

Precision (PREC):

$$PREC = \frac{TP}{TP + FP}$$

PREC is computed as the number of correct positive predictions divided by the total number of positive predictions.

Recall (REC):

$$REC = \frac{TP}{TP + FN}$$

REC is computed as the number of correct positive predictions divided by the total number of positives. It represents the relevant patients that have been correctly detected, it is also called Sensitivity or true positive rate (TPR).

F-Measure:

$$F\text{-}Measure := 2 * \frac{PREC * REC}{PREC + REC}$$

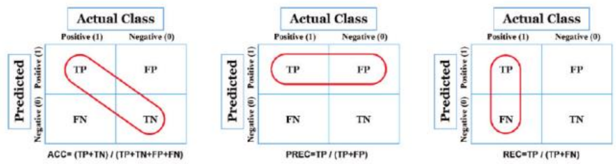

Figure 4 Performance metrics: Accuracy, Precision, Recall.

F-Measure called also F-score, is a harmonic mean of precision and recall, it provides the quality of prediction.

ROC – AUC Area:

AUC – ROC curve is a performance measurement for the classification problems at various threshold settings. ROC is a probability curve and AUC represents the degree or measure of separability. It tells how much the model is capable of distinguishing between classes. If the value of AUC is high, the model predicts classes indicated by 0 as value 0 and classes indicated by 1 as value 1. By analogy, when the value of the AUC is high, the model is more efficient and therefore we can distinguish patients with disease and without disease.

There are other metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), but generally are used in regression problems. Therefore, this comparative study will rely on the performance metrics explained above due to the dataset and algorithms used categorized in classification problems. Also, the same metrics are used to evaluate the quality of our ontology model.

In the next section, we present the result obtained from the classifiers using Weka and Protégé software.

5 Results and Discussion

In this section, we present the result obtained from the evaluation of classifiers used in this research including the result and statistics of the ontology classifier.

This study is based on a set of criteria, on the one hand, no method applied for feature selection or performance improvement for a fair comparison of the performance of classification algorithms, on the other hand, we used two modes test: cross-validation 10 times and percentage split (split 66.0% train,

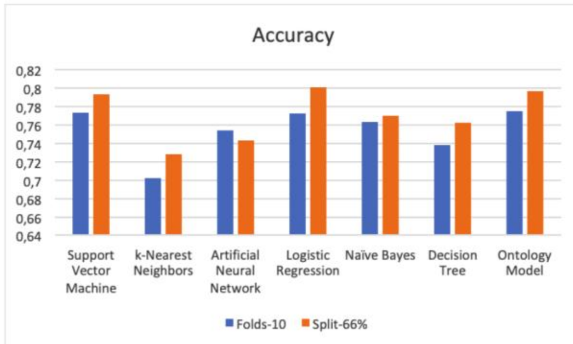

Figure 6 Comparison results of accuracy.

– **Precision**

The ontology classifier has the highest Precision of 81.2% for both test modes. Followed by Naïve Bayes and ANN. More details are shown in Table 4 and Figure 7.

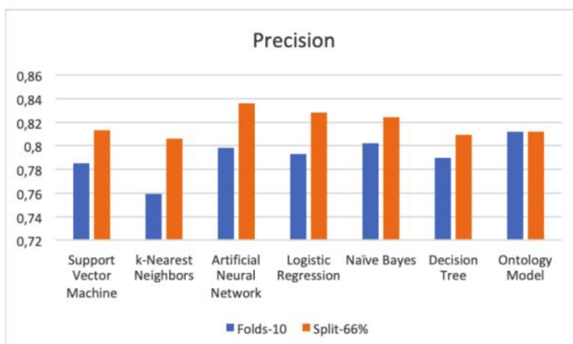

Figure 7 Comparison results of precision.

– **ROC area**

Table 4 and Figure 10 show that Logistic Regression, Naïve Bayes, and Ontology have the better value of the ROC Area.

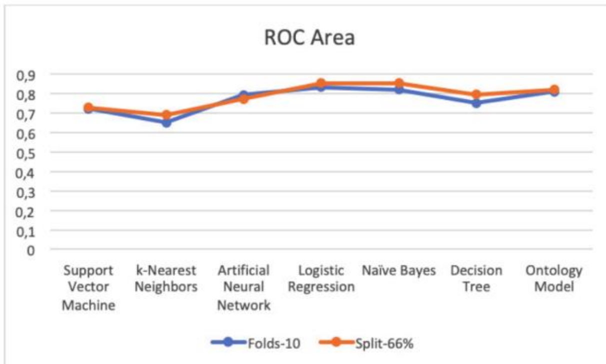

Figure 10 Comparison results of ROC area.

Table 4 Statistics of the experimental results for ML and ontology classifiers

	Accuracy		Precision		Recall		F-Measure		ROC Area	
	Folds-10	Split-66%	Folds-10	Split-66%	Folds-10	Split-66%	Folds-10	Split-66%	Folds-10	Split-66%
SVM	0,773	0,813	0,785	0,813	0,898	0,904	0,838	0,856	0,720	0,729
KNN	0,702	0,806	0,759	0,806	0,794	0,792	0,776	0,799	0,650	0,691
ANN	0,754	0,836	0,798	0,836	0,832	0,775	0,815	0,805	0,793	0,772
LR	0,772	0,828	0,793	0,828	0,880	0,893	0,834	0,859	0,832	0,855
NB	0,763	0,824	0,802	0,824	0,844	0,843	0,823	0,833	0,819	0,854
DT	0,738	0,809	0,790	0,809	0,814	0,854	0,802	0,831	0,751	0,796
Ontology	0,775	0,812	0,812	0,812	0,867	0,909	0,838	0,858	0,808	0,819

Discussion

In our measurements, we used two test mode options, and we noticed that the percentage split was exceeded in the cross-validation test mode due to the small data mass, for this we will base by following on a cross-validation 10 times.

leads us to a new search field that we suggest and encourage researchers to contribute and create new ideas in the same context, to give more results and comparison, for the purpose of prediction, recommendation, or make a decision, etc.

From our side, we look forward to enhancing this comparative study by applying new approaches to integrate rules of machine learning with the ontology classification method, we also intend to use regression machine learning algorithms

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